

PART 1

1. Analytical Approach

$$D(x) = (x - x_0)^2 + (f(x) - y_0)^2$$

given $f(x) = x^2 + 5$ and in all cases $y_0 = 0$

$$\begin{aligned} D(x) &= (x - x_0)^2 + (x^2 + 5 - 0)^2 \\ &= (x - x_0)^2 + (x^2 + 5)^2 \end{aligned}$$

then set the derivative $D'(x) = 0$

$$\therefore D'(x) = 2(x - x_0) + 2(x^2 + 5)(2x) = 0$$

$$\Rightarrow x - x_0 + 2x^3 + 10x = 0$$

$$\Rightarrow 2x^3 + 11x - x_0 = 0$$

a. point $(0, 0) \Rightarrow x_0 = 0$

$$\therefore D'(x) = 2x^3 + 11x = 0$$

$$x(2x^2 + 11) = 0$$

$$\therefore \text{REAL roots: } x = 0$$

$$\Rightarrow f(0) = 5$$

$$\therefore \text{point on curve: } (0, 5)$$

$$\therefore \text{shortest distance: } d = \sqrt{(0-0)^2 + (5-0)^2} = 5$$

b. point $(-4, 0)$

$$\therefore D'(x) = 2x^3 + 11x + 4 = 0$$

$$\Rightarrow \text{using calculator: } x \approx -0.3503$$

$$\Rightarrow f(-0.3503) \approx 5.1227$$

$$\therefore \text{point on curve: } (-0.3503, 5.1227)$$

$$\therefore \text{shortest distance: } d = \sqrt{(-0.3503 + 4)^2 + (5.1227 - 0)^2} \approx 6.2847$$

c. point $(-8, 0)$

$$\therefore D'(x) = 2x^3 + 11x + 8 = 0$$

$$\Rightarrow \text{using calculator: } x \approx -0.6698$$

$$\Rightarrow f(-0.6698) \approx 5.4486$$

$$\therefore \text{point on curve: } (-0.6698, 5.4486)$$

$$\therefore \text{shortest distance: } d = \sqrt{(-0.6698 + 8)^2 + (5.4486 - 0)^2} \approx 9.1382$$

d. point $(2, 0)$

$$\therefore D'(x) = 2x^3 + 11x - 2 = 0$$

$$\Rightarrow \text{using calculator: } x \approx 0.1793$$

$$\Rightarrow f(0.1793) \approx 5.0322$$

$$\therefore \text{point on curve: } (0.1793, 5.0322)$$

$$\therefore \text{shortest distance: } d = \sqrt{(0.1793 - 2)^2 + (5.0322 - 0)^2} \approx 5.3524$$

e. point $(6, 0)$

$$\therefore D'(x) = 2x^3 + 11x - 6 = 0$$

$$\Rightarrow \text{using calculator: } x \approx 0.5186$$

$$\Rightarrow f(0.5186) \approx 5.2689$$

$$\therefore \text{point on curve: } (0.5186, 5.2689)$$

$$\therefore \text{shortest distance: } d = \sqrt{(0.5186 - 6)^2 + (5.2689 - 0)^2} \approx 7.6166$$

2. numerical method A - Newton-Raphson method

$$D(x) = (x-x_0)^2 + (x^2+5)^2 \quad \text{* same as before *$$

$$\begin{aligned} D'(x) &= 2(x-x_0) + 2(x^2+5)(2x) \\ &= 2x - 2x_0 + 4x^3 + 20x \\ &= 4x^3 + 22x - 2x_0 \end{aligned}$$

$$D''(x) = 12x^2 + 22$$

$$\begin{aligned} \text{iterative formula: } x_{k+1} &= x_k - \frac{D'(x_k)}{D''(x_k)} \\ &= x_k - \frac{4x_k^3 + 22x_k - 2x_0}{12x_k^2 + 22} \end{aligned}$$

a. point (0,0)

↳ iteration 1: guess $x_0 = 0$

$$D'(x_0) = D'(0) = 0$$

$$D''(x_0) = D''(0) = 0$$

$$x_1 = x_0 - \frac{D'(x_0)}{D''(x_0)} = 0$$

since it already converged in one step $x=0$ is the correct value

$$f(0) = 5$$

$$\begin{aligned} d &= \sqrt{(x-x_0)^2 + (f(x)-y_0)^2} \\ &= \sqrt{(0-0)^2 + (5-0)^2} \\ &= 5 \end{aligned}$$

b. point (-4,0)

↳ iteration 1: guess $x_0 = -2.0$

$$D'(x_0) = D'(-2) = 4(-2)^3 + 22(-2) - 2(-4) = -68$$

$$D''(x_0) = D''(-2) = 12(-2)^2 + 22 = 70$$

$$x_1 = x_0 + \frac{D'(x_0)}{D''(x_0)} = -2 - \frac{-68}{70} = -1.028571$$

↳ iteration 2: $x_1 = -1.028571$

$$D'(-1.028571) = -26.974026$$

$$D''(-1.028571) = 34.695510$$

$$x_2 = x_1 + \frac{D'(x_1)}{D''(x_1)} = -1.028571 - \frac{-26.974026}{34.695510} = -0.481489$$

↳ iteration 3: $x_2 = -0.481489$

$$D'(-0.481489) = -3.036063$$

$$D''(-0.481489) = 24.782012$$

$$x_3 = x_2 + \frac{D'(x_2)}{D''(x_2)} = -0.481489 + \frac{-3.036063}{24.782012} = -0.358849$$

$$f(x_3) = (-0.358849)^2 + 5 = 5.12852$$

$$\begin{aligned} d &= \sqrt{(x-x_0)^2 + (f(x)-y_0)^2} \\ &= \sqrt{(-0.358849+4)^2 + (5.12852-0)^2} \\ &= 6.289854 \end{aligned}$$

c. point $(-8, 0)$

↳ guess $x_0 = -4.0$

$$x_1 = x_0 - \frac{D'(x_0)}{D''(x_0)} = -4 - \frac{-328}{214} \doteq -2.467290$$

$$x_2 = -2.467290 - \frac{-98.3752}{95.0689} \doteq -1.432478$$

$$x_3 = -1.432478 - \frac{-27.270570}{46.622584} \doteq -0.847558$$

$$f(x_3) = (-0.847558)^2 + 5 = 5.718355$$

$$d = \sqrt{(-0.847558 + 8)^2 + (5.718355 - 0)^2} \\ = 9.157347$$

d. point $(2, 0)$:

↳ guess $x_0 = 1.0$

$$x_1 = x_0 - \frac{D'(x_0)}{D''(x_0)} = 1 - \frac{22}{34} \doteq 0.352941$$

$$x_2 = 0.352941 - \frac{3.94025}{23.4948} \doteq 0.185220$$

$$x_3 = 0.185223 - \frac{0.100326}{22.411684} \doteq 0.180747$$

$$f(x_3) = (0.180747)^2 + 5 = 5.032669$$

$$d = \sqrt{(0.180747 - 2)^2 + (5.032669 - 0)^2} \\ = 5.351396$$

e. point $(6, 0)$

↳ guess $x_0 = 3.0$

$$x_1 = x_0 - \frac{D'(x_0)}{D''(x_0)} = 3 - \frac{162}{130} \doteq 1.753846$$

$$x_2 = 1.753846 - \frac{48.162916}{58.911712} \doteq 0.936262$$

$$x_3 = 0.936262 - \frac{11.880632}{32.519044} \doteq 0.570919$$

$$f(x_3) = (0.570919)^2 + 5 = 5.325949$$

$$d = \sqrt{(0.570919 - 6)^2 + (5.325949 - 0)^2} \\ = 7.605304$$

3. numerical approach 8 - golden section search

$$\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618034$$

$$r = 2 - \varphi \approx 0.381966$$

$$D(x) = (x-x_0)^2 + (x^2+5-y_0)^2$$

a. point (0,0)

$$D(x) = x^2 + (x^2+5)^2$$

↳ iteration 1:

$$\text{initial bracket: } [a, b] = [-15, 15]$$

$$x_1 = a + r(b-a) = -15 + 0.381966(15+15) = -3.541020$$

$$x_2 = b - r(b-a) = 15 - 0.381966(15+15) = 3.541020$$

$$D(x_1) = D(x_2) = 321.328$$

$$\therefore \text{tie rule} \rightarrow a = -3.541020$$

↳ iteration 2:

$$\text{bracket: } [-3.541020, 15]$$

$$x_1 = 3.541020 \quad (\text{kept from last iteration})$$

$$x_2 = 7.917961$$

$$D'(x_1) = 321.328$$

$$D'(x_2) = 4645.1705$$

$$D'(x_1) < D'(x_2) \rightarrow [a, b] = [-3.541020, 7.917961]$$

$$x = \frac{a+b}{2} = \frac{-3.541020 + 7.917961}{2} = 2.188471$$

$$y = x^2 + 5 = (2.188471)^2 + 5 = 9.789405$$

$$d = \sqrt{(2.188471-0)^2 + (9.789405-0)^2} \\ = 10.03$$

b. point (-4,0)

$$D(x) = (x+4)^2 + (x^2+5)^2$$

↳ iteration 1

$$\text{initial bracket: } [-19, 11]$$

$$x_1 = -19 + r(30) = -7.5410$$

$$x_2 = 11 - r(30) = -0.4590$$

$$D(x_1) = (-7.5410 + 4)^2 + ((-7.5410)^2 + 5)^2 = 3840.0617$$

$$D(x_2) = (-0.4590 + 4)^2 + ((-0.4590)^2 + 5)^2 = 39.6898$$

$$D(x_1) > D(x_2) \rightarrow \text{keep } [x_1, b] = [-7.5410, 11]$$

↳ iteration 2: [-7.5410, 11]

$$x_1 = -0.4590$$

$$x_2 = 3.9180$$

$$D(x_1) = 39.6898$$

$$D(x_2) = 476.8335$$

$$\left. \begin{array}{l} D(x_1) = 39.6898 \\ D(x_2) = 476.8335 \end{array} \right\} D(x_1) < D(x_2) \rightarrow \text{keep } [a, x_2] \rightarrow [-7.5410, 3.9180]$$

$$x = \frac{a+b}{2} = \frac{-7.5410 + 3.9180}{2} = -1.8115$$

$$f(x) = (-1.8115)^2 + 5 = 8.2816$$

$$d = \sqrt{(-1.8115 + 4)^2 + (8.2816)^2} = 8.5659$$

c. point $(-8, 0)$

$$D(x) = (x+8)^2 + (x^2+5)^2$$

↳ iteration 1

$$\text{initial bracket: } [-23, 7]$$

$$x_1 = -23 + r(30) = -11.5410$$

$$x_2 = 7 - r(30) = -4.4590$$

$$D(x_1) = (-11.5410 + 8)^2 + ((-11.5410)^2 + 5)^2 = 19110.4341$$

$$D(x_2) = (-4.4590 + 8)^2 + ((-4.4590)^2 + 5)^2 = 631.6779$$

$$D(x_1) > D(x_2) \rightarrow \text{keep } [x_1, b] = [-11.5410, 7]$$

↳ iteration 2 : $[-11.5410, 7]$

$$x_1 = -4.4590$$

$$x_2 = -0.0820$$

$$\left. \begin{array}{l} D(x_1) = 631.6779 \\ D(x_2) = 87.7615 \end{array} \right\} D(x_1) > D(x_2) \rightarrow [x_1, b] = [-4.4590, 7]$$

$$x = \frac{a+b}{2} = \frac{-4.4590 + 7}{2} = 1.2705$$

$$f(x) = (1.2705)^2 + 5 = 6.6142$$

$$d = \sqrt{(1.2705 + 8)^2 + (6.6142)^2} = 11.3881$$

d. point $(2, 0)$

$$D(x) = (x-2)^2 + (x^2+5)^2$$

↳ iteration 1

$$\text{initial bracket: } [-13, 17]$$

$$x_1 = -13 + r(30) = -1.5410$$

$$x_2 = 17 - r(30) = 5.5410$$

$$D(x_1) = (-1.5410 - 2)^2 + ((-1.5410)^2 + 5)^2 = 66.9256$$

$$D(x_2) = (5.5410 - 2)^2 + ((5.5410)^2 + 5)^2 = 1287.2358$$

$$D(x_1) < D(x_2) \rightarrow \text{keep } [a, x_2] = [-13, 5.5410]$$

↳ iteration 2 : $[-13, 5.5410]$

$$x_1 = -5.9180$$

$$x_2 = 1.5410$$

$$\left. \begin{array}{l} D(x_1) = 1664.4753 \\ D(x_2) = 66.9256 \end{array} \right\} D(x_1) > D(x_2) \rightarrow \text{keep } [x_1, b] = [-5.9180, 5.5410]$$

$$x = \frac{a+b}{2} = \frac{-5.9180 + 5.5410}{2} = -0.1885$$

$$f(x) = (-0.1885)^2 + 5 = 5.0355$$

$$d = \sqrt{(-0.1885 - 2)^2 + (5.0355)^2} = 5.4905$$

e. point (6,0)

$$D(x) = (x-6)^2 + (x^2+5)^2$$

↳ iteration 1

initial bracket : $[-9, 21]$

$$x_1 = -9 + r(30) = 2.4590$$

$$x_2 = 21 - r(30) = 9.5410$$

$$D(x_1) = (2.4590 - 6)^2 + ((2.4590)^2 + 5)^2 \doteq 134.5658$$

$$D(x_2) = (9.5410 - 6)^2 + ((9.5410)^2 + 5)^2 \doteq 9234.5026$$

$$D(x_1) < D(x_2) \rightarrow \text{keep } [a, x_2] = [-9, 9.5410]$$

↳ iteration 2 : $[-9, 9.5410]$

$$x_1 = -9 + r(9.5410 + 9) = -1.9180$$

$$x_2 = 2.4590$$

$$D(x_1) \doteq 138.0117 \quad \left\{ \begin{array}{l} D(x_1) > D(x_2) \rightarrow \text{keep } [x_1, b] = [-1.9180, 9.5410] \end{array} \right.$$

$$D(x_2) \doteq 134.5658$$

$$x = \frac{a+b}{2} = \frac{-1.9180 + 9.5410}{2} \doteq 3.8115$$

$$f(x) = 19.5278$$

$$d = \sqrt{(3.8115 - 6)^2 + (19.5278)^2} \doteq 19.6500$$

PART II

1. Analytic approach

dataset: $(0, 0.5)$, $(1, 1.5)$, $(2, 3.5)$, $(3, 7.5)$

a. linear model ($y = mx + b$)

$$MSE_{\text{line}}(m, b) = \frac{1}{4} \sum_{i=1}^4 (y_i - (mx_i + b))^2$$

$$\frac{\partial MSE}{\partial m} = -\frac{2}{4} \sum_{i=1}^4 x_i (y_i - mx_i - b) = 0 \rightarrow m \sum x_i^2 + b \sum x_i = \sum x_i y_i$$

$$\frac{\partial MSE}{\partial b} = -\frac{2}{4} \sum_{i=1}^4 (y_i - mx_i - b) = 0 \rightarrow m \sum x_i + 4b = \sum y_i$$

$$\sum x_i = 0 + 1 + 2 + 3 = 6$$

$$\sum y_i = 0.5 + 1.5 + 3.5 + 7.5 = 13$$

$$\sum x_i^2 = 0^2 + 1^2 + 2^2 + 3^2 = 14$$

$$\sum x_i y_i = 0(0.5) + 1(1.5) + 2(3.5) + 3(7.5) = 31$$

$$\therefore 14m + 6b = 31 \quad (1)$$

$$6m + 4b = 13 \quad (2)$$

$$\text{rearrange (2): } b = 3.25 - 1.5m$$

$$\text{sub (2) in (1): } 14m + 6(3.25 - 1.5m) = 31$$

$$14m + 19.5 - 9m = 31$$

$$m = 2.3$$

$$\therefore b = 3.25 - 1.5(2.3)$$

$$b = -0.2$$

$$\therefore \text{optimal linear model: } y = 2.3x - 0.2$$

	$\hat{y}_i = 2.3x_i - 0.2$	residuals $(y_i - \hat{y}_i)$
$x_1 = 0$	$\hat{y}_1 = -0.2$	0.7
$x_2 = 1$	$\hat{y}_2 = 2.1$	-0.6
$x_3 = 2$	$\hat{y}_3 = 4.4$	-0.9
$x_4 = 3$	$\hat{y}_4 = 6.7$	0.8

$$MSE_{\text{line}} = \frac{1}{4} ((0.7)^2 + (-0.6)^2 + (-0.9)^2 + (0.8)^2) = 0.5750$$

b. parabolic model ($y = ax^2 + bx + c$)

$$MSE_{\text{parabola}}(a, b, c) = \frac{1}{4} \sum_{i=1}^4 (y_i - (ax_i^2 + bx_i + c))^2$$

$$\frac{\partial MSE}{\partial a} = -\frac{2}{n} \sum x_i^2 r_i = 0$$

$$\frac{\partial MSE}{\partial b} = -\frac{2}{n} \sum x_i r_i = 0$$

$$\frac{\partial MSE}{\partial c} = -\frac{2}{n} \sum r_i = 0$$

$$\sum x_i^3 = 0^3 + 1^3 + 2^3 + 3^3 = 36$$

$$\sum x_i^4 = 0^4 + 1^4 + 2^4 + 3^4 = 98$$

$$\sum x_i^2 y_i = 0^2(0.5) + 1^2(1.5) + 2^2(3.5) + 3^2(7.5) = 83$$

$$98a + 36b + 14c = 83$$

$$36a + 14b + 6c = 31$$

$$14a + 6b + 4c = 13$$

$$\left. \begin{array}{l} \text{using} \\ \text{elimination} \\ \text{\&} \\ \text{substitution} \end{array} \right\} \rightarrow \begin{array}{l} a = 0.75 \\ b = 0.05 \\ c = 0.55 \end{array}$$

∴ best-fit parabola $\rightarrow y = 0.75x^2 + 0.05x + 0.55$

	$\hat{y}_i = 0.75x_i^2 + 0.05x_i + 0.55$	residuals ($y_i - \hat{y}_i$)
$x_1 = 0 \rightarrow$	$\hat{y}_1 = 0.55 \rightarrow$	-0.05
$x_2 = 1 \rightarrow$	$\hat{y}_2 = 1.35 \rightarrow$	0.15
$x_3 = 2 \rightarrow$	$\hat{y}_3 = 3.65 \rightarrow$	-0.15
$x_4 = 3 \rightarrow$	$\hat{y}_4 = 7.45 \rightarrow$	0.05

$$\text{MSE}_{\text{parabola}} = \frac{(-0.05)^2 + (0.15)^2 + (-0.15)^2 + (0.05)^2}{4} = 0.0125$$

2. numerical approach : Newton-Raphson

Newton update formula : $\theta_{k+1} \leftarrow \theta_k - \frac{\frac{\partial \text{MSE}}{\partial \theta}}{\frac{\partial^2 \text{MSE}}{\partial \theta^2}}$

a. linear model

$$\frac{\partial \text{MSE}}{\partial m} = -\frac{2}{n} \sum x_i (y_i - mx_i - b)$$

$$\frac{\partial^2 \text{MSE}}{\partial m^2} = \frac{2}{n} \sum x_i^2$$

$$\frac{\partial \text{MSE}}{\partial b} = -\frac{2}{n} \sum y_i$$

$$\frac{\partial^2 \text{MSE}}{\partial b^2} = 2$$

start with $m=0$, $b=0$

↳ at those values all y-values are zero

↳ residual errors are (0.5, 1.5, 3.5, 7.5)

↳ try updating m :

$$\frac{\partial \text{MSE}}{\partial m} = -\frac{2}{4} [0(0.5) + 1(1.5) + 2(3.5) + 3(7.5)] = -15.5$$

$$\frac{\partial^2 \text{MSE}}{\partial m^2} = \frac{2}{4} (0^2 + 1^2 + 2^2 + 3^2) = 7$$

↳ apply Newton update : $m = 0 - \frac{-15.5}{7} = 2.214286$

∴ new iteration: $m = 2.214286$, $b = 0$

↳ recalculated MSE = 0.589286

↳ recomputed residual errors : (0.5, -0.714286, -0.928571, 0.857143)

↳ try updating b :

$$\frac{\partial \text{MSE}}{\partial b} = 0.142857$$

$$\frac{\partial^2 \text{MSE}}{\partial b^2} = 2$$

↳ apply Newton update : $b = 0 - \frac{0.142857}{2} = -0.071429$

∴ new iteration: $m = 2.214286$, $b = -0.071429$

↳ recalculated MSE = 0.584134

(repeat)

b. parabolic model (where $r_i = y_i - (ax_i^2 + bx_i + c)$)

$$\frac{\partial \text{MSE}}{\partial a} = -\frac{2}{n} \sum x_i^2 r_i = 0$$

$$\frac{\partial \text{MSE}}{\partial b} = -\frac{2}{n} \sum x_i r_i = 0$$

$$\frac{\partial \text{MSE}}{\partial c} = -\frac{2}{n} \sum r_i = 0$$

$$\frac{\partial^2 \text{MSE}}{\partial a^2} = \frac{2}{n} \sum x_i^4$$

$$\frac{\partial^2 \text{MSE}}{\partial b^2} = \frac{2}{n} \sum x_i^2$$

$$\frac{\partial^2 \text{MSE}}{\partial c^2} = 2$$

start at $a=0, b=0, c=0$

↳ initial MSE = 17.75

↳ try updating a:

$$\frac{\partial \text{MSE}}{\partial a} = -41.5$$

$$\frac{\partial^2 \text{MSE}}{\partial a^2} = 49$$

↳ apply Newton's update: $a = 0 - \frac{-41.5}{49} = 0.846939$

new iteration: $a = 0.846939, b = c = 0$

↳ recalculated MSE: 0.176020

↳ recalculated residual errors: (0.5, 0.616618, 0.039359, -0.231778)

↳ try updating b:

$$\frac{\partial \text{MSE}}{\partial b} = -0.255102$$

$$\frac{\partial^2 \text{MSE}}{\partial b^2} = 7$$

↳ apply Newton's update: $b = 0 - \frac{-0.255102}{7} = 0.036443$

new iteration: $a = 0.846939, b = 0.036443, c = 0$

↳ MSE = 0.171372

↳ residual error: (0.5, 0.616618, 0.039359, -0.231778)

↳ try updating c:

$$\frac{\partial \text{MSE}}{\partial c} = -0.462099$$

$$\frac{\partial^2 \text{MSE}}{\partial c^2} = 2$$

↳ apply Newton's update: $c = 0 - \frac{-0.462099}{2} = 0.231050$

new iteration: $a = 0.846939, b = 0.036443, c = 0.231050$

↳ MSE = 0.117988

(repeat)